

Project: Build panorama generation software for store shelves (Submitted by Prerana.B(prerana.bora120@gmail.com))

Problem statement:

Develop a system that stitches multiple overlapping images of a retail store into a single seamless panorama while preserving accurate shelf geometry and product placement.

Rules

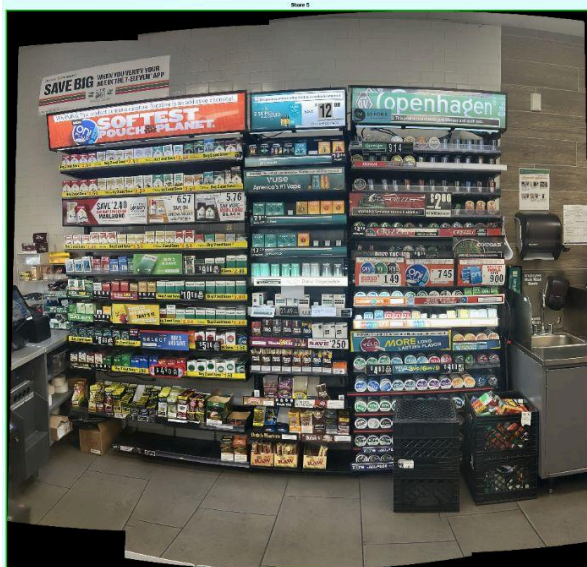
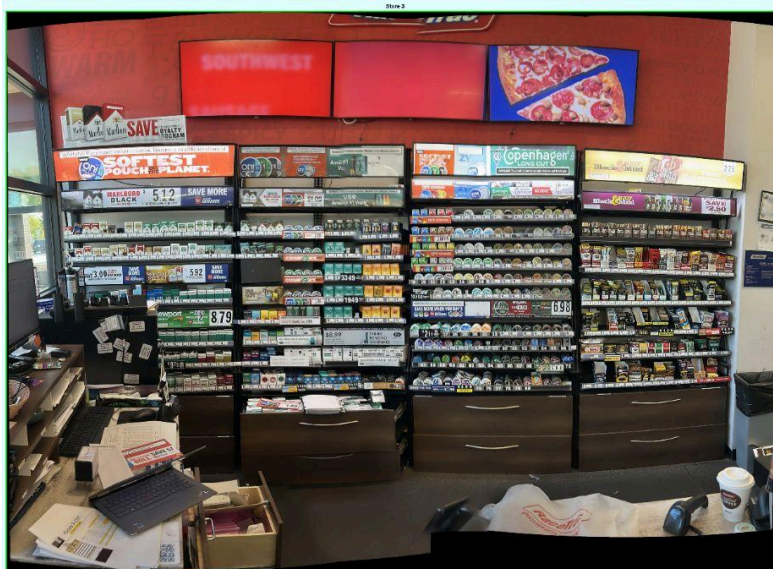
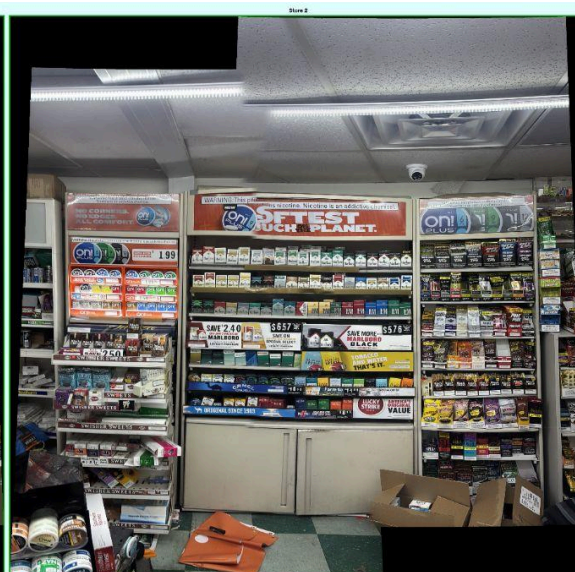
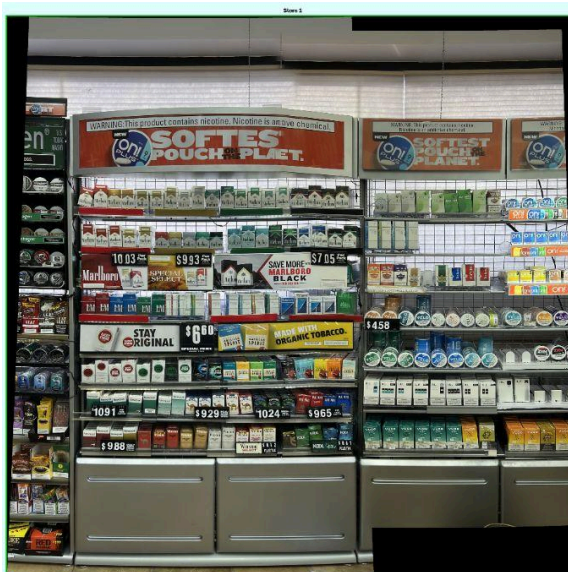
- Generate one panorama per store.
- Include all visible fixture regions.
- Do not duplicate products in overlapping areas.
- Preserve straight shelf alignment.
- Minimize ghosting, warping, tearing, and visible seams.
- Keep product packaging and price tags readable.
- Ensure the panorama is suitable for downstream retail analysis.

Approaches experimented:

Approach 1: OpenCV cv2.Stitcher + SIFT Fallback

- cv2.Stitcher_PANORAMA → handles full pipeline in one call → fastest baseline
- If fails → retry at 50% resolution → stabilizes matching on high-res images
- If still fails → SIFT + FLANN + Lowe's ratio test → filters false matches from repetitive shelf patterns 0.75
- RANSAC homography → rejects outliers → alpha blend → crop black borders 5.0px

Results:

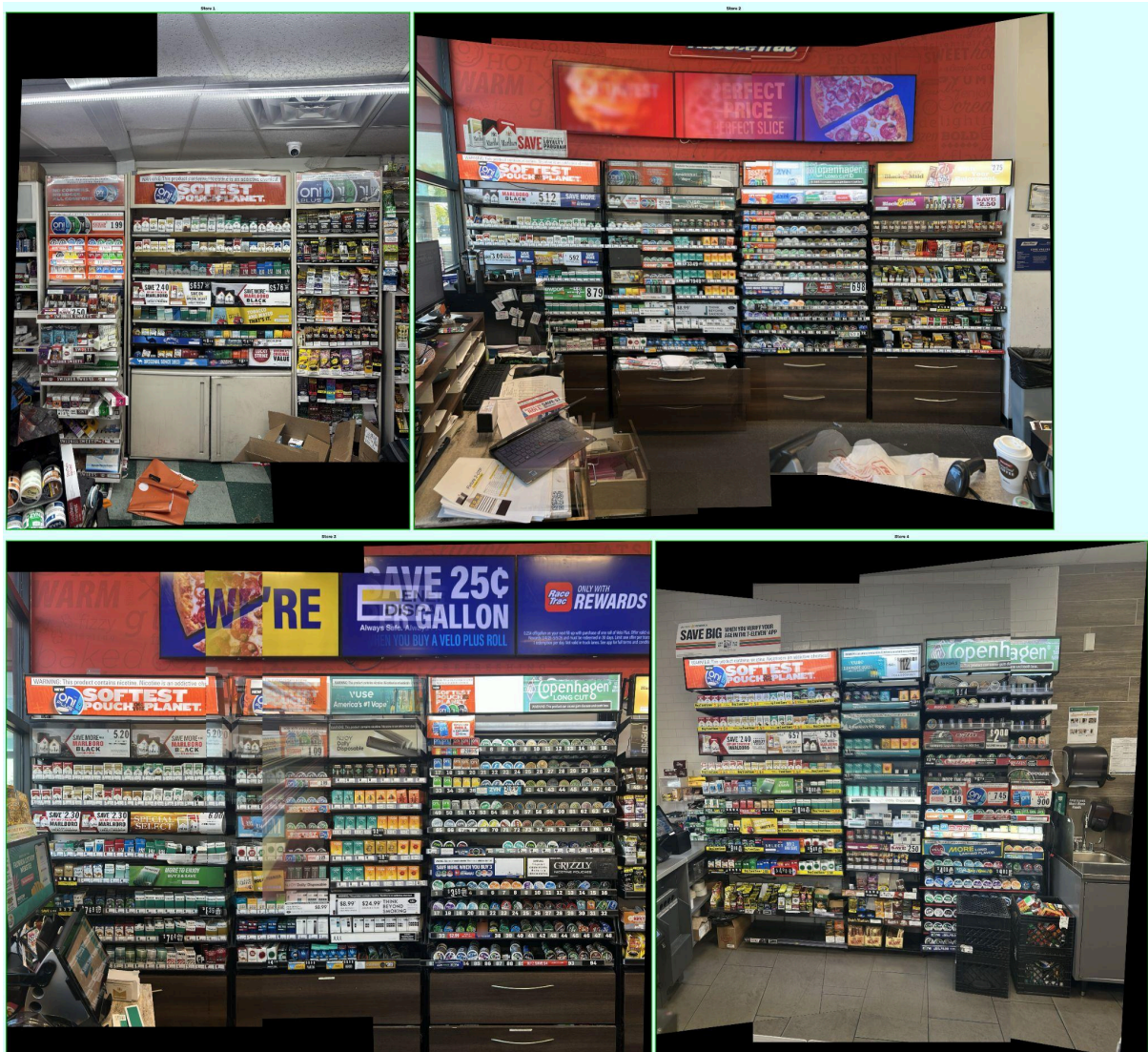


Observations: **cv2.Stitcher** + SIFT fallback failed for **Store 1** and **Store 4** (multi-image cases) → repetitive shelf textures generate too many outlier matches → RANSAC cannot find stable homography across 3+ images.

Approach 2: SuperPoint + LightGlue

- SuperPoint → deep learned detector → robust on repetitive shelf textures
- LightGlue transformer matcher → global context matching → fewer false matches than nearest-neighbour
- RANSAC homography → tighter threshold justified by high-quality matches
- Sequential stitching left to right → alpha blend overlap → crop borders
- **Limitation:** Computationally heavy and process is slow.

Results:



Observations: For Store 1 it is complete failed because 4K resolution with repetitive shelf textures produced too few reliable matches (6 inliers from 49) → compounding bad homographies across 3+ images caused panorama to explode to

21628×12653px.

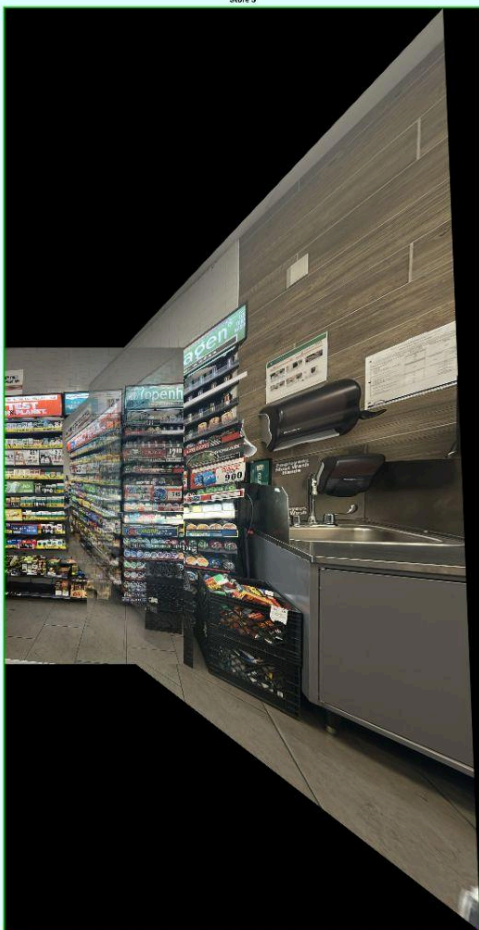
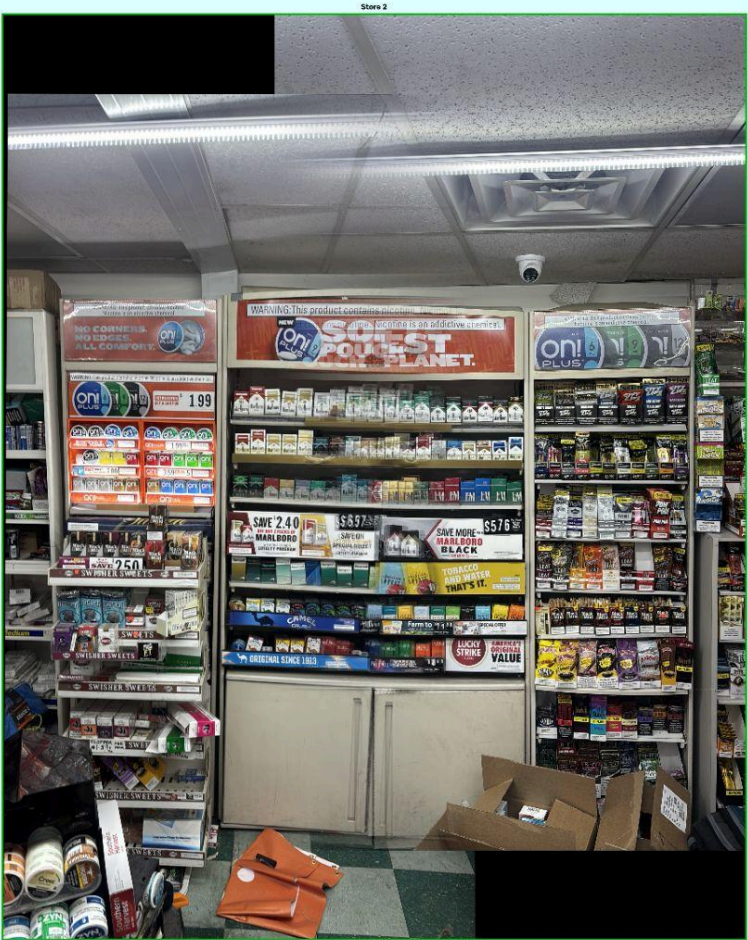
Store 4 is distorted.

Store 5 lost resolution.

Approach 3: LoFTR

- LoFTR → closer to retail store environment `pretrained='indoor'`
- Dense matcher → operates on full image grid → no separate detect + describe → directly outputs matches end-to-end
- Resize to → fixed size required for transformer attention → scale keypoints back to original resolution for accurate warping(`480, 640`)
- Filter → removes low-quality matches in textureless shelf areas `confidence >= 0.5`
- RANSAC → high iterations handle near-duplicate points from dense matching `3.0px maxIters=10000`
- Warp + horizontal gradient alpha blend + crop borders
- Limitations: Fixed resize loses fine detail on 4K images → slow on CPU

Result:



Observations: Failed completely for Store 1, 3, 4, 5 → fixed resize of 4K images causes severe resolution loss → dense matching finds geometrically inconsistent correspondences despite high confidence → very low RANSAC inliers → produces heavily distorted panoramas. (480×640)

Final Approach:

RetailGlue : Semantic Product-Level Image Stitching for Retail Shelf Panoramas

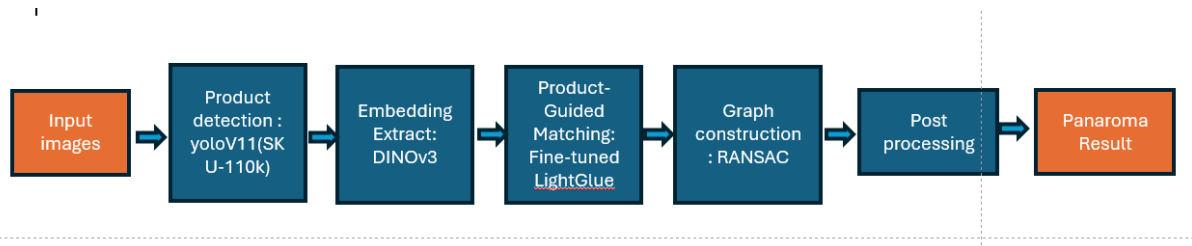
(<https://github.com/rempeople/RetailGlue/tree/main>)

Pipeline

1. Load all store images sorted by filename → consistent left-to-right sequence
2. For each consecutive image pair → run YOLOv11 → detect product/shelf bounding boxes
3. Crop detected product regions → extract DINOv3 embeddings per crop → builds semantic identity per product
4. Use YOLO box centers as semantic anchor points + SuperPoint/LoFTR keypoints as geometric anchor points
5. Match anchor points between image pairs → filter by confidence threshold → retain only reliable matches
6. Match DINOv3 embeddings across image pairs → confirms same product is matched → prevents false matches on visually similar products
7. Compute homography matrix **H** via RANSAC → **H is the projective transform that maps pixels from source image onto destination image plane** 3×3
8. Validate **H** → check inlier ratio → reject if too few inliers → unreliable transform
9. Project 4 corners of **source image** (image being warped) through **H** → compute output canvas size → apply translation to shift coordinates positive $T @ H$

10. Warp **source image** (next image in sequence) onto canvas using → aligns it with current panorama `cv2.warpPerspective`
11. Build overlap mask → use YOLO box IoU + DINOv2 similarity → detect and remove duplicate products in overlap zone
12. Apply horizontal gradient alpha blend in overlap region → smooth seam transition
13. Crop black borders → clean canvas edges
14. Accumulate stitched result as new panorama → becomes source for next iteration
15. Repeat steps 2-14 for each next image
16. Save final panorama per store

Diagram:

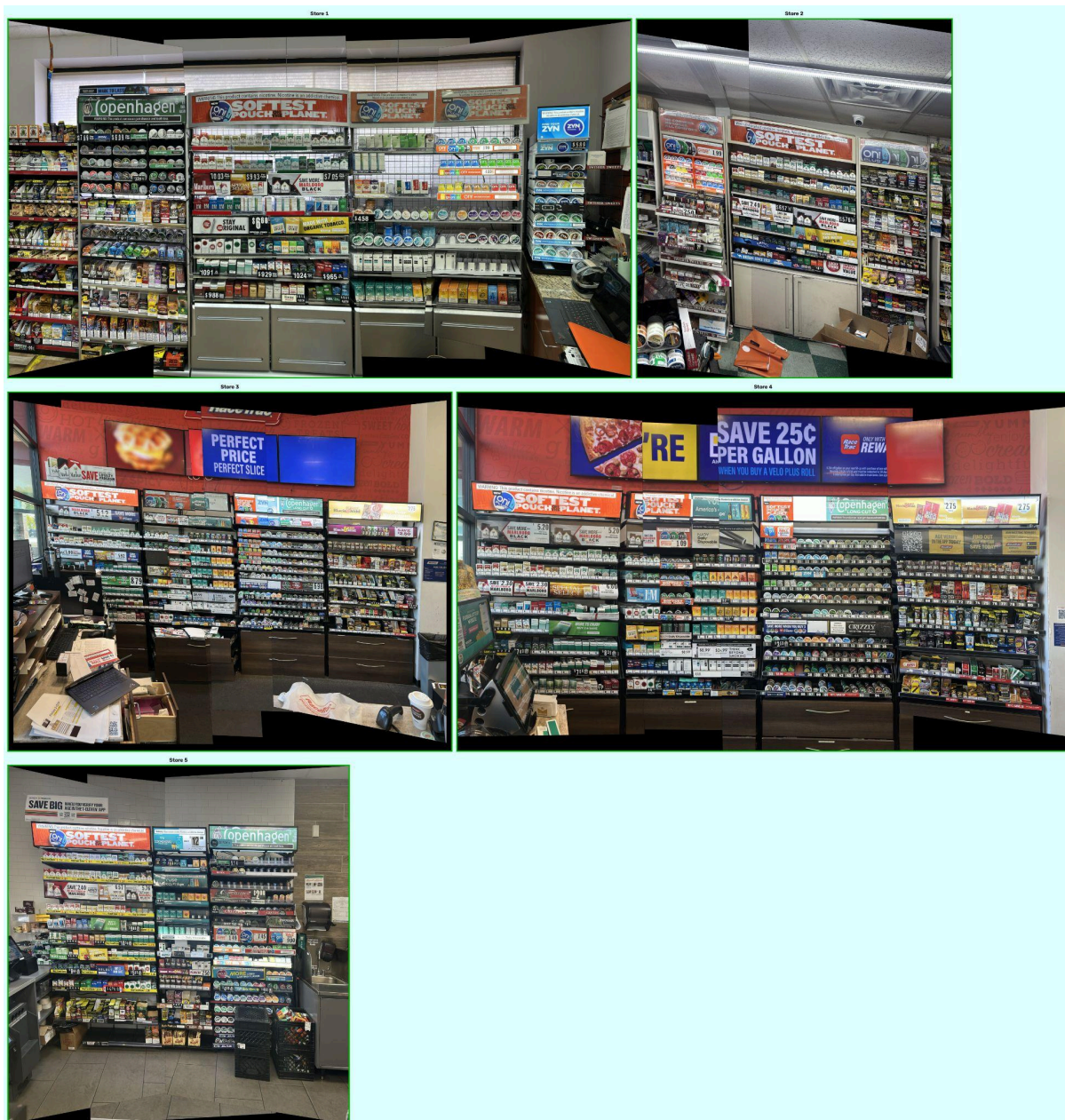


End-to-End Stitching Quality on RemRetail100_v2:

Pipeline / Matcher	Dim.	Acc.	Prec.	Rec.	F1
<i>Local feature matching pipelines (baselines)</i>					
SuperPoint + LightGlueStick	256	80.2	90.2	87.0	87.9
<i>RetailGlue</i>					
DINOv3 ViT-B/16 + LightGlue	768	80.7	89.4	87.7	88.4

RetailGlue consistently outperforms the baselines in runtime, completing medium-sized (8-image) sequences in **2.34 s** compared to **9.76 s** for DISK + LightGlue and **15.82 s** for RoMav2.

Results:



Note: Using DINOv3 ViT-S/16

Observations: RetailGlue produces panoramas that closely match the provided reference panoramas. Preserved high-resolution details and avoided duplicate product instances. The reconstructed panoramas are geometrically consistent across all evaluated stores; only Store 2 exhibits a slight tilt in the final stitched output.

Solution to fix Store 2 tilt stitched panorama:

1. RetailGlue faced Homography Estimation Issue

With only 2 images, fewer point combinations available to compute homography → USAC-MAGSAC accepts it as geometrically valid but tilted.

2. Sparse Keypoints = Less Geometric Constraint

Only product centers used → too few RANSAC inliers → small uncorrected angular error causes visible tilt.

Why cv2. Stitcher + SIFT Gives Better Result for stitching =>2 images:

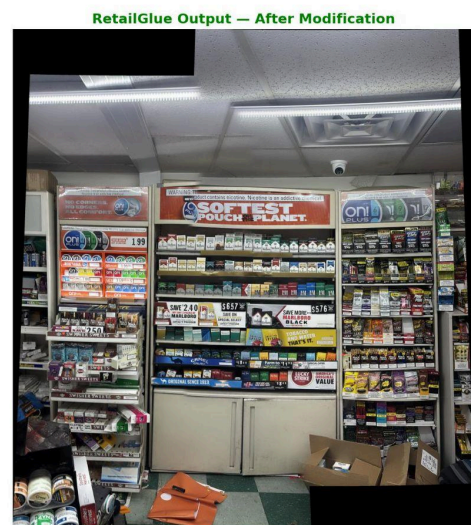
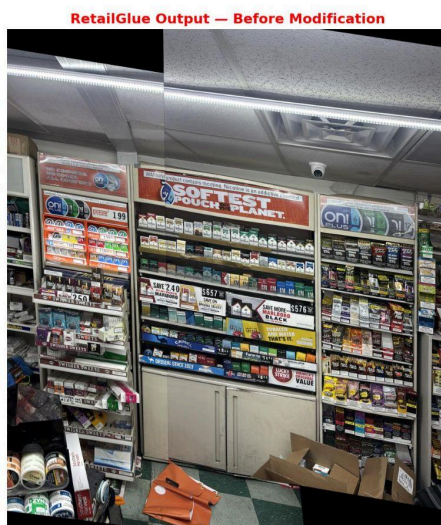
- SIFT finds **thousands of well-distributed keypoints** on 4K images
- cv2. Stitcher runs **bundle adjustment + wave correction** → tilt corrected automatically
- RetailGlue's strength is **repetitive/identical products** where SIFT gets confused by ambiguous matches

Fixed Approach added in stitcher.py for stitching 2 images:

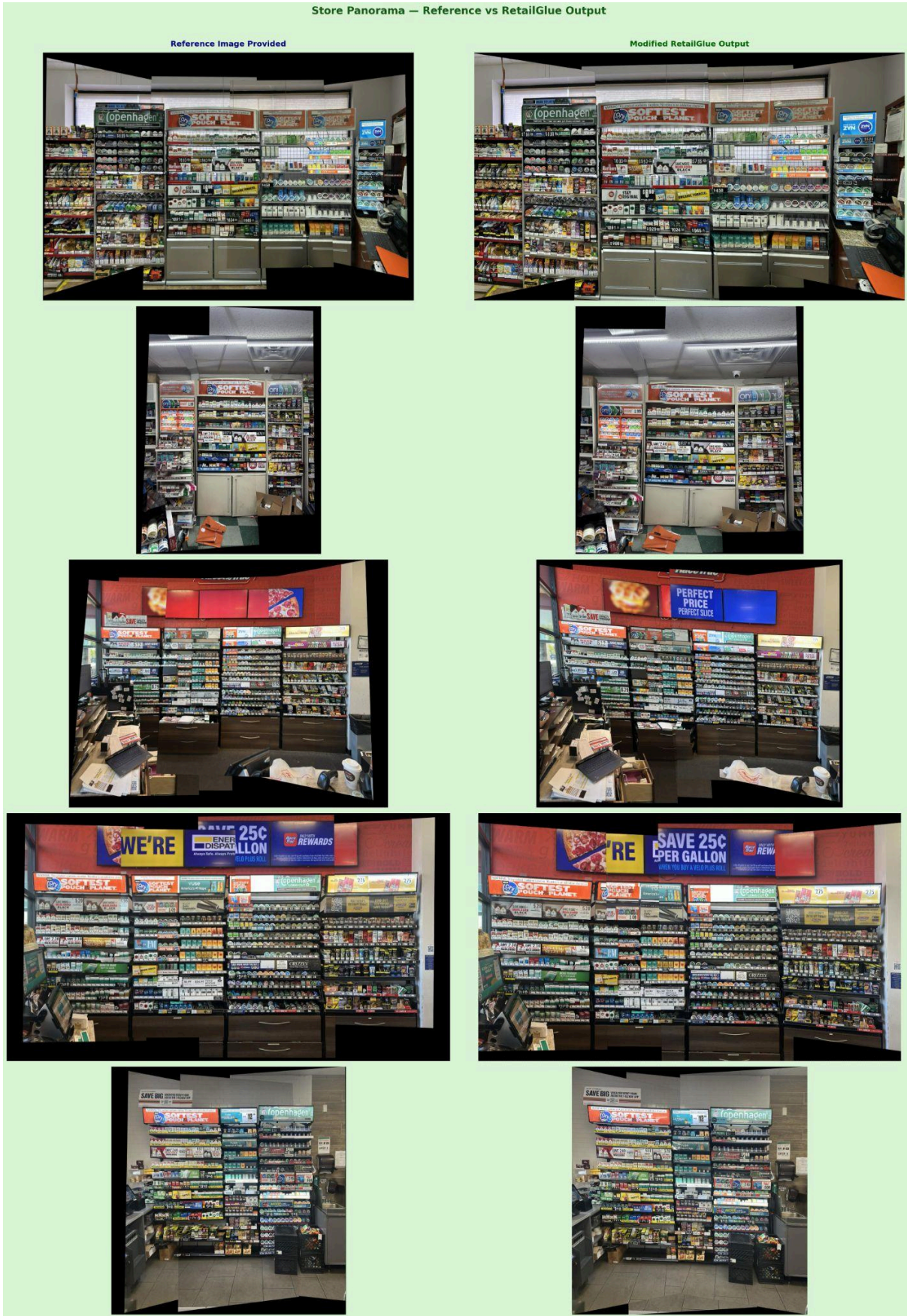
1. **2-image pipeline bypass** — When only 2 images are detected, the existing tilt-prone graph pipeline is skipped and a dedicated fix path runs instead
2. **Hybrid stitching** — RetailGlue's product matches compute the homography; then takes over for warping, brightness correction & smooth blending cv2.detail
3. **Tilt correction** — Homography is broken into possible rotations; the one with least tilt (closest to identity) is picked, then straightens the final `panoramawaveCorrect(HORIZ)`
4. **Fallback to cv2. Stitcher** — If hybrid stitching fails due to too few matches or bad homography, (SIFT-based, more keypoints) automatically takes over cv2.Stitcher
5. **Black border removal** — After warping, cleans up any black edges left around the `panorama_crop_black_borders()`

Results:

Store 2 — Panorama Stitching Comparison



Comparison with provided Reference image:



Personal Contribution: Progressive RANSAC Adapted from Jia et al. (2024) into RetailGlue

From Jia et al. 2024([Semantic Aware Stitching for Panorama](#))(No code available), the core iterative plane-extraction logic is adapted for a RetailGlue.

RetailGlue originally used a single RANSAC to compute one global homography transform between overlapping shelf images.

Motivation for adapting Progressive RANSAC:

- A **single RANSAC** tries to fit **one global transform** to ALL correspondences across ALL depth planes
- This produces a **noisy, averaged similarity matrix** — not truly accurate for any single plane
- Result: **misalignment, ghosting, blurring** in stitched output

What Jia et al. proposed:

Step	Action
1	Run RANSAC with lenient threshold (0.225) → remove only gross outliers
2	Run RANSAC with strict threshold (0.2) → extract one plane's inliers → compute that plane's centroid
3	Remove remaining points spatially close to that centroid
4	Repeat Steps 2 & 3 until inlier ratio < 0.30
5	Each inlier set → compute its own similarity transformation matrix

Personal Adaptations on basis of RetailGlue's requirement:

1. **Adopted Progressive RANSAC concept from Jia et al. 2024** — replaced RetailGlue's original single RANSAC with an iterative plane-extraction approach, so each depth plane (shelf row) gets its own accurate similarity transform instead of one noisy global transform.
2. **Adapted plane removal criterion** — changed from Jia's centroid-proximity removal to **reprojection error-based removal**.
3. **Added resolution-independent normalization** — coordinates normalized to making thresholds camera-agnostic across different store deployments. `[0, 1]`
4. **Introduced semantic pre-filters** — added **DINOv3 embedding similarity** and **LightGlue confidence** filters before RANSAC to reduce false matches caused by visually repetitive retail products.
5. **Added plane quality scoring** — each extracted plane ranked by , ensuring the dominant shelf plane drives the final warp. `score = (n_inliers × mean_embedding_similarity) / mean_reprojection_error`
6. Added this whole script in `/RetailGlue/retailglue/stitching/progressive_ransac.py` and called it from `/Users/testuser/Desktop/store_panorama_stiching/RetailGlue/retailglue/stitching/stitcher.py`

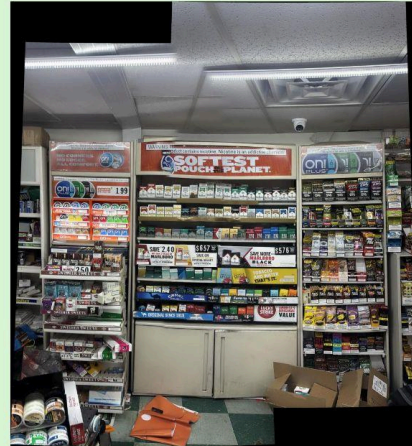
Results:

Store Panorama — All Outputs

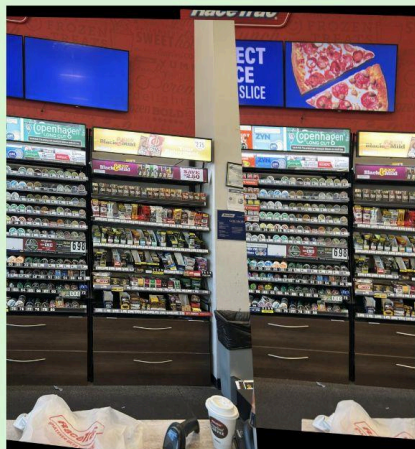
Store 1



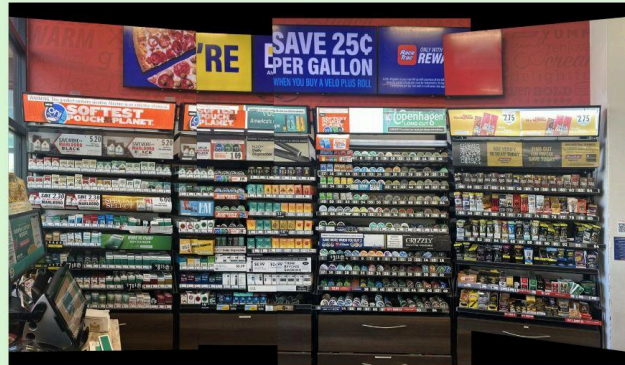
Store 2



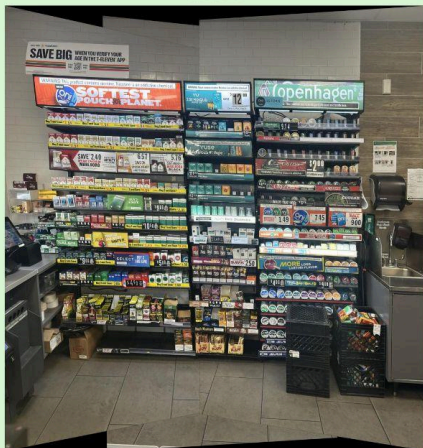
Store 3



Store 4



Store 5



Observations:

- With Progressive RANSAC, Store 3 panorama stitched differently compared to single RANSAC output
- The result is not necessarily better — the dominant plane selected may not be the true shelf plane for Store 3
- Suggests thresholds may need store-specific tuning
- Further diagnosis needed — visualizing selected inliers per plane would clarify which plane is driving the warp

References

1. Oztuner, A., Celik, A., Yalciner, I. S., & Calap, S. (2024). *RetailGlue: Semantic Product-Level Image Stitching for Retail Shelf Panoramas*. REM People. <https://github.com/rempeople/RetailGlue/tree/main>
2. Jia, Y., Li, Z., Zhang, L., Song, B., & Song, R. (2024). Semantic Aware Stitching for Panorama. *Sensors*, 24(11), 3512. <https://doi.org/10.3390/s24113512>
3. Anthropic. (2025). *Claude (claude-sonnet-4-5)*. Retrieved from <https://claude.ai>
4. Connected Papers. (2025). *Connected Papers — Find and explore academic papers*. Retrieved from <https://www.connectedpapers.com>
5. Google Scholar. (2025). *Google Scholar*. Retrieved from <https://scholar.google.com>
6. Tuli, M., Kamali, K., & Lindell, D. B. (2024). *Generative Panoramic Image Stitching*. Retrieved from <https://arxiv.org/abs/2406.11anmar>
7. Feng, Y. (2024). Research on Image Stitching Based on an Improved LightGlue Algorithm. Retrieved from <https://doi.org/10.3390/s24113512>